

The term System-on-Chip refers to the chip housed in your mobile device, which pretty much functions as a complete computer. Now while desktop systems have a separate processor, separate GPU (except for those systems with integrated processors), separate controllers to handle the peripheral devices and so on, that is not the case with mobile devices such as tablets and cellphones. All of the discrete components seen on the desktop need to be integrated within one single chip and that implementation is called as the System-on-Chip (SoC).

So the SoC is basically an integrated circuit, with all the parts of a computer on one single chip. The SoC is what defines a mobile device and it contains within it the processor cores, the graphics chips, the cellular radios, mobile RAM, image processors, audio/video decoders and so on. The consolidation of all these seemingly separate-function parts on a single chip also helps in keeping the power requirements lower. These days you get SoCs which can easily beat the desktop system configs from say five years back. And with every successive generation the SoCs are getting even more impressive.

For the desktop systems we have Intel who design and manufacture their own chips. But in the mobile space ARM Holdings, a British company, is the bigger name. Unlike Intel, ARM Holdings makes the processor refer-

